Fullstack Development

Part 2: The battle of the meta frameworks

> Next.js vs Astro vs TanStack Start

What meta-frameworks do

- **File-Based Routing**
  - Files and folders define the URL structure of your app.
- **Server-Side Rendering (SSR)**
  - Builds HTML on a server to make pages load fast and work well with search engines.
- **Static Site Generation (SSG)**
  - Builds HTML at build time, so the server doesn't have to do any work when a user visits.

What meta-frameworks do

- **Data Fetching**
  - Provides a way to fetch data from a database or API and pass it to your components.
- **Server Actions / RPC**
  - Lets you call server-side code from the client without writing your own API endpoints.
- **Optimization**: Speeds up images, scripts, and code automatically

Elevator pitches

- `Astro`: A content-first web framework that ships zero JavaScript by default.
- `Next.js`: A production framework for React, architected server-first around Server Components.
- `TanStack Start`: A production framework for React, architected client-first around a full SPA router — with SSR built in.

Modern rendering taxonomy

- **Static Site Generation (SSG) / Static Prerendering / Partial Prerendering (PPR)**
  - Build-time HTML, no server needed.
- **Server-Side Rendering (SSR)**
  - HTML is built on the server per request.
  - Optional streaming
- **SSR + Hydration / Isomorphic Rendering**
  - HTML is built on the server, then the client takes over and makes it interactive.
- **Client-Side Rendering (CSR)**
  - HTML is built in the browser, no server rendering.

Summary

Name	When it runs	Where	Delivery
SSG	Build time	Server	One complete HTML file
SSR	Request time	Server	One response (or streamed)
Isomorphic	Request time	Server + Client	One response, then rehydrated
CSR	Request time	Client	Blank shell, then client-rendered

Summary (with frameworks)

Name	Astro	Next.js	TanStack Start
SSG	Astro Component	Server Component	*
SSR	Server Island	Server Component	*
Isomorphic	Client Island	Client Component	Any components
CSR	*	*	*

> * = possible, but not the default and/or recommended way.

Todo app

The todo app

- Astro
  - `git clone https://github.com/fullstack-69/ls-astro.git`
- Next.js
  - `git clone https://github.com/fullstack-69/ls-nextjs.git`
- TanStack Start
  - `git clone https://github.com/fullstack-69/ls-tanstack-start.git`
- Same UI (Pico CSS), same database, same features.
- Only the **architecture** differs.

Three parts, three timestamps

Component	What it does	Question it answers
`Nav`	prints a timestamp	When was the shell made?
`Greeting`	reads `user-agent` + prints a timestamp	Was this per-request?
`Todo`	list + delete (interactive)	Where does the data enter?

> If a timestamp never changes on refresh → it was **baked at build time**.

Shared data layer

`src/db/index.ts` — identical in all three projects

```
export async function getTodos() {
  await ensureTodosTable();
  await sleep(DB_LATENCY); // artificial latency, so we can SEE the loading states
  const result = await client.execute(
    "SELECT id, todoText FROM todos ORDER BY rowid ASC",
  );
  return result.rows.map((row) => toTodo(row));
}

export async function deleteTodo(id: number) { ... }
```

- libSQL / SQLite (`file:app.db`).
- **Server-only code**. It must never reach the browser bundle.

Diagram

https://link.excalidraw.com/l/9PltHIQHZMD/4zVYOlJ54Ee

DX comparison

> From a perspective of a developer who knows React and SPA frameworks.

Astro — two familiar things, kept separate

- The Astro component is closer to a classic template language (Handlebars, EJS, Pug) than to React.
  - `---` fence = data fetching. Template below = HTML output. No hooks, no lifecycle.
  - Fits my intuition immediately because it is the old-school model.
- The React island is **100% ordinary React**
  - It is a fully separate render tree with its own bundle.
  - No new React concepts. `useState`, `useEffect`, callbacks — all work as expected.

Next.js — familiar syntax, unfamiliar rules

- Next repurposes **ordinary-looking React syntax** to mean something environment-dependent.
  - A string literal (`'use client'` / `'use server'`) is intercepted by the compiler and silently changes what the file is allowed to do and where it runs.
- It looks identical to the React I know. It behaves differently.
  - You cannot `import` a Server Component inside a `'use client'` file.
  - No `useState` in a Server Component.

TanStack Start — explicit over implicit

- No magic string. No compiler rewriting the meaning of a component file.
- The server/client boundary is drawn by **which function you're inside**, not by a hidden annotation on the file.
- Writing React component code is the **same everywhere**. The only difference is whether you call a server function or not.
- `TanStack Router` and `Query` are libraries SPA developers already have muscle memory for — Start reuses them rather than replacing them.

References

> This part is AI-generated from my codebase. Use it for reference, not as a tutorial.

Astro

Astro component

`ls-astro/src/components/Greeting.astro`

```
---
import { UAParser } from "ua-parser-js";

const uaHeader = Astro.request.headers.get("user-agent") || "";
const result = new UAParser(uaHeader).getResult();
const timeStr = new Date().toLocaleTimeString();
---

<article class="card">
  <h1>Hello, {result.browser.name}!</h1>
  <p>Generated at: {timeStr}</p>
</article>
```

Astro component (anatomy)

- **Component script** — between the `---` fences.
  - Plain TS/JS, runs **only on the server**, never shipped to the browser.
  - Can `import` server-only modules freely (db, `node:fs`, secrets).
- **Component template** — below the fences.
  - HTML + `{expressions}`.
  - Renders **once**, to a string. No re-render, no hooks, no state.
- Output: pure HTML. **Zero JS**.

Astro page

`ls-astro/src/pages/index.astro`

```
---
import Layout from "@components/Layout.astro";
import Nav from "@components/Nav.astro";
import Greeting from "@components/Greeting.astro";
import TodoReact from "@components/Todo.tsx";
---

<Layout>
  <Nav />
  <Greeting server:defer>
    <article slot="fallback"><p>Loading <span aria-busy="true"></span></p></article>
  </Greeting>
  <TodoReact client:load />
</Layout>
```

Astro page (routing)

- **File-based routing** — anything in `src/pages/` becomes a route.
  - `src/pages/index.astro` → `/`
  - `src/pages/blog/[slug].astro` → `/blog/:slug`
- A page is just an `.astro` component that returns a full document.
- `src/components/Layout.astro` is the html shell; children arrive via `<slot />`.

```
<body class="container">
  <slot />
</body>
```

Island architecture

> The page is a **sea of static HTML**. Interactivity lives in isolated **islands**.

Each island hydrates independently. No island → no JS.

Two kinds of islands

```
<Nav />                          <!-- 1. plain: static HTML, 0 KB JS -->
<Greeting server:defer />        <!-- 2. server island: HTML fetched at request time -->
<TodoReact client:load />        <!-- 3. client island: React, hydrated in browser -->
```

	Renders where	JS to browser
`<Nav />`	server, at build	none
`server:defer`	server, per request	none
`client:load`	server + browser	React + component

Client islands — hydration directives

Directive	Hydrates when
`client:load`	immediately on page load
`client:idle`	browser is idle
`client:visible`	island scrolls into viewport
`client:media`	CSS media query matches
`client:only`	never SSR'd — client render only

> Islands are the **only** JS in the bundle: `dist/client/_astro/Todo.*.js`.

Server islands

`ls-astro/src/pages/index.astro`

```
<Greeting server:defer>
  <article slot="fallback">
    <h1>Greeting</h1>
    <p>
      Loading <span aria-busy="true"></span>
    </p>
  </article>
</Greeting>
```

- Page HTML ships **immediately** with the `fallback`.
- The island is then fetched from `/_server-islands/Greeting` and swapped in.
- The static page can sit on a CDN; only the island hits the origin.

Astro rendering mode

`ls-astro/astro.config.mjs`

```
export default defineConfig({
  adapter: node({ mode: "standalone" }),
  integrations: [react()],
});
```

- `output: "static"` **is the default** — pages are prerendered to HTML at build.
- The `adapter` adds a server, for islands / actions / SSR pages.

Opting out, per page

`ls-astro/src/pages/some-page.astro`

```
---
export const prerender = false;   // this page now renders per request
---
```

Mode	When
prerendered (default)	content is the same for everyone
`prerender = false`	page depends on the request (auth, personalization)

> Astro makes you opt **in** to the server. Next.js and Start make you opt **out**.

Proof: what actually got built

```
dist/
├── client/
│   ├── index.html           ← the whole page, prerendered
│   └── _astro/Todo.*.js     ← the only JS shipped
└── server/
    └── chunks/Greeting.mjs ← the server island, rendered on demand
```

- `Nav`'s timestamp is **frozen** in `index.html` (build time).
- `Greeting`'s timestamp changes on **every request**.
- `Todo`'s data arrives **after** hydration.

Astro Actions

ls-astro/src/actions/index.ts

```ts
import { defineAction } from "astro:actions";
import { z } from "astro/zod";
import { getTodos, deleteTodo } from "../db/index.ts";

export const server = {
  getTodos: defineAction({
    handler: async () => await getTodos(),
  }),
  deleteTodo: defineAction({
    input: z.object({ id: z.number() }), // runtime validation
    handler: async ({ id }) => {
      await deleteTodo(id);
      return null;
    },
  }),
};
```

Calling an Action from an island

`ls-astro/src/components/Todo.tsx`

```
import { actions } from "astro:actions";

const { data, error } = await actions.getTodos();
if (error) {
  console.error(error);
  return;
}
setTodos(data);

// delete
await actions.deleteTodo({ id: todo.id });
```

Calling an Action from an island

- Type-safe RPC — no `fetch`, no URL, no manual JSON.
- Returns `{ data, error }`; errors are **values**, not exceptions.
- The `import` looks local, but it compiles to a POST to `/_actions/...`.

Astro summary

- Static HTML is the **default**; everything else is opt-in.
- `client:*` → client island (React ships).
- `server:defer` → server island (no JS ships).
- `actions` → the typed client → server boundary.
- Mental model: **an HTML document that occasionally contains an app**.

Next.js

App Router

```
ls-nextjs/src/app/
├── layout.tsx    → root layout (wraps every route)
├── page.tsx      → the route "/"
└── favicon.ico
```

- **Folders define routes; files define UI.**
- Reserved filenames: `page`, `layout`, `loading`, `error`, `not-found`, `route`, `template`.
- `app/blog/[slug]/page.tsx` → `/blog/:slug`

Root layout

ls-nextjs/src/app/layout.tsx

```tsx
export const metadata: Metadata = {
  title: "Next.js",
  description: "A Next.js app",
};

export default function RootLayout({
  children,
}: {
  children: React.ReactNode;
}) {
  return (
    <html lang="en">
      <body className="container">{children}</body>
    </html>
  );
}
```

- Owns `<html>` / `<body>`. Persists across navigation (does not re-render).

The page

ls-nextjs/src/app/page.tsx

```tsx
export default function Home() {
  return (
    <>
      <Nav />
      <Suspense
        fallback={
          <article>
            <h1>Greeting</h1>
            <p>Loading…</p>
          </article>
        }
      >
        <Greeting />
      </Suspense>
      <Todo />
    </>
  );
}
```

The page

- Compare with `index.astro` — same three components, same shape.
- `<Suspense>` replaces Astro's `server:defer` + `slot="fallback"`.

Server Component

`ls-nextjs/src/components/Greeting.tsx`

```tsx
import { headers } from "next/headers";

const Greeting: FC = async () => {
  const headersList = await headers(); // server-only API
  const uaHeader = headersList.get("user-agent") || "";
  const browser = new UAParser(uaHeader).getResult().browser.name;
  const timeStr = new Date().toLocaleTimeString();

  return (
    <article className="card">
      <h1>Hello, {browser}!</h1>
      <p>Generated at: {timeStr}</p>
    </article>
  );
};
```

Server Component (rules)

- **The default**. No directive needed — every component in `app/` is a Server Component.
- Can be `async` and `await` directly in the component body.
- Can touch the database, secrets, `headers()`, `cookies()`.
- Ships **zero JS** — the browser receives the rendered output, not the code.
- Cannot use `useState`, `useEffect`, `onClick`, or any browser API.

> This is the direct analogue of the `---` fence in an `.astro` file.

Client Component

ls-nextjs/src/components/Todo.tsx

```
"use client";
import { useEffect, useState, useTransition } from "react";
import { actionDeleteTodo, actionGetTodos } from "@/actions";

const TodoComponent: FC = () => {
  const [todos, setTodos] = useState<Todo[]>([]);
  const [loading, startTransition] = useTransition();

  useEffect(() => { startTransition(fetchTodos); }, []);
  ...
};
```

Client Component (rules)

- `"use client"` at the top of the file marks the **boundary**, not the file alone.
  - Everything imported *below* it also becomes client code.
- Still **server-rendered once** (for the HTML), then **hydrated** in the browser.
- Gets hooks, event handlers, browser APIs.
- Astro's `client:load` = Next's `"use client"` — but Next has **no** `client:visible`; hydration is not scheduled per-component.

Server Actions

`ls-nextjs/src/actions/index.ts`

```ts
"use server"; // Marks ALL exports in this file as Server Actions

import { deleteTodo, getTodos } from "@/db";

export async function actionGetTodos() {
  return await getTodos();
}

export async function actionDeleteTodo(id: number) {
  await deleteTodo(id);
  return null;
}
```

Calling a Server Action

`ls-nextjs/src/components/Todo.tsx`

```
const data = await actionGetTodos(); // looks like a local call...
setTodos(data);

await actionDeleteTodo(todo.id); // ...is actually a POST to the server
```

- The bundler replaces the function body with an **RPC stub + an ID**.
- Throws on error → wrap in `try/catch` (Astro returns `{ data, error }` instead).
- ⚠️ No built-in input validation: `id: number` is a *compile-time* claim only. Astro's `input: z.object(...)` and TanStack's `.validator()` do this for you.

Partial Prerendering (PPR)

> One route = a **static shell** delivered instantly + **dynamic holes** streamed in.

`ls-nextjs/next.config.ts`

```ts
const nextConfig: NextConfig = {
  reactCompiler: true,
  cacheComponents: true, // Enables 'use cache' behavior
};
```

- Previously `experimental.ppr`; in Next 16 it is driven by `cacheComponents`.

PPR — the static half

`ls-nextjs/src/components/Nav.tsx`

```
const Nav: FC = async () => {
  "use cache"; // → prerendered into the shell
  const timeStr = new Date().toLocaleTimeString();
  return (
    <nav>
      ...<a href="#NA">Generated at: {timeStr}</a>...
    </nav>
  );
};
```

- `"use cache"` = "this output is not per-request, bake it."
- Its timestamp is **frozen at build**, exactly like Astro's `Nav.astro`.

PPR — the dynamic half

`ls-nextjs/src/app/page.tsx`

```
<Suspense fallback={<p>Loading…</p>}>
  <Greeting /> {/* calls headers() → cannot be prerendered */}
</Suspense>
```

- `headers()` makes `Greeting` **dynamic**, so it must not block the shell.
- Next prerenders the page **with the fallback in place**, then streams the real HTML into the hole.
- Forget the `<Suspense>` and the build fails — the boundary is the contract.

PPR vs. Astro server islands

	Astro	Next.js
Static shell	`.astro` page, prerendered	route shell, prerendered
Dynamic hole	`<Greeting server:defer>`	`<Suspense><Greeting/></Suspense>`
Placeholder	`slot="fallback"`	`fallback={...}`
Delivery	second HTTP request	streamed in the same response
Marked by	directive on the tag	*inferred* from dynamic API usage

Next.js summary

- Server is the default; the client is opt-in via `"use client"`.
- One language, one component model, one mental model — React everywhere.
- Static / dynamic is **inferred** from what your code touches, not declared.
- Mental model: **an app that happens to render on the server first**.

TanStack Start

File-based routing

```
ls-tanstack-start/src/routes/
├── __root.tsx                  → the shell (html/head/body)
├── index.tsx                   → /
├── ssr_prerendered.tsx         → /ssr_prerendered
└── ssr_awaited.tsx             → /ssr_awaited
```

- Files → routes, but **compiled into a typed route tree**: `src/routeTree.gen.ts`.
- Generated automatically by the Vite plugin. **Never edit it.**

The root route

ls-tanstack-start/src/routes/__root.tsx

```
export const Route = createRootRoute({
  head: () => ({
    meta: [{ title: "TanStack Start" }, { charSet: "utf-8" }],
    links: [{ rel: "stylesheet", href: picoStyles }],
  }),
  shellComponent: RootDocument,
});

function RootDocument({ children }: { children: React.ReactNode }) {
  return (
    <html lang="en">
      <head>
        <HeadContent />
      </head>
      <body>
        {children}
        <Scripts />
      </body>
    </html>
  );
}
```

A route

`ls-tanstack-start/src/routes/index.tsx`

```
export const Route = createFileRoute("/")({
  component: Home,
  loader: async () => {
    const nav = await getNavTime();
    const greeting = await getGreeting();
    return { nav, greeting };
  },
});
```

- A route is a **declarative object**: a path, a component, and its data.
- Not a default export — the router imports `Route` by convention.

The route's component

`ls-tanstack-start/src/routes/index.tsx`

```
function Home() {
  const state = Route.useLoaderData(); // fully typed, no generics needed

  return (
    <main className="container">
      <Nav timeStr={state.nav.timeStr} />
      <Greeting
        browser={state.greeting.browser}
        timeStr={state.greeting.timeStr}
      />
      <Todo />
    </main>
  );
}
```

Loaders

- A route **declares its own data** in `loader`.
  - Runs on the server during SSR, and on the client during SPA navigation.
- `Route.useLoaderData()` is typed from the loader's return — no manual types.
- Contrast:
  - Next: each Server Component fetches its own data.
  - Astro: the page frontmatter fetches.
  - Start: **the route** fetches, then passes down as props.

Typed links

`ls-tanstack-start/src/components/Nav.tsx`

```
import { Link, useLocation } from "@tanstack/react-router";

<Link
  to="/ssr_prerendered"
  className={isActive("/ssr_prerendered") ? "active" : ""}
>
  SSR Prerendered
</Link>;
```

- `to` is checked against the generated route tree — a typo is a **type error**.
- Client-side navigation (SPA), with `defaultPreload: "intent"` prefetching on hover.

Isomorphic components

`ls-tanstack-start/src/components/Greeting.tsx`

```tsx
interface GreetingProps {
  browser: string;
  timeStr: string;
}

const Greeting: FC<GreetingProps> = ({ browser, timeStr }) => (
  <article className="card">
    <h1>Hello, {browser}!</h1>
    <p>Generated at: {timeStr}</p>
  </article>
);
```

> **No** `"use client"`. **No** `"use server"`. **No** `client:load`.

Isomorphic — what it means

- There is **one** kind of component. It runs on the server (SSR) *and* in the browser (hydration).
- Every component ships to the client. There is no "server-only component".
- So the server/client boundary is **not** the component — it is the **server function**.

Framework	Boundary drawn at
Astro	the island tag (`client:load`)
Next.js	the file (`"use client"`)
Start	the **function call** (`createServerFn`)

Server functions

`ls-tanstack-start/src/actions/index.ts`

```ts
import { createServerFn } from "@tanstack/react-start";
import { getRequestHeaders } from "@tanstack/react-start/server";

export const getGreeting = createServerFn({ method: "GET" }).handler(
  async () => {
    const header = getRequestHeaders();
    const ua = header.get("user-agent") ?? "";
    return {
      browser: new UAParser(ua).getResult().browser.name || "Unknown",
      timeStr: new Date().toLocaleTimeString(),
    };
  },
);
```

- Explicit `method` — `GET` is cacheable, `POST` is a mutation.

Server functions — validation

`ls-tanstack-start/src/actions/index.ts`

```
const deleteTodoSchema = z.object({ id: z.number() });

export const actionDeleteTodo = createServerFn({ method: "POST" })
  .validator(deleteTodoSchema) // runtime-validated input
  .handler(async ({ data }) => {
    await deleteTodo(data.id);
    return null;
  });
```

- Builder chain: `createServerFn` → `.middleware()` → `.validator()` → `.handler()`.
- The `.handler` body is **stripped from the client bundle**; the call becomes an RPC.

Calling a server function

`ls-tanstack-start/src/components/Todo.tsx`

```
import { useServerFn } from "@tanstack/react-start";

const getTodos = useServerFn(actionGetTodos);
const deleteTodo = useServerFn(actionDeleteTodo);

const data = await getTodos();
await deleteTodo({ data: { id: todo.id } }); // note the { data: ... } envelope
```

- Direct call works too; `useServerFn` additionally integrates with the router (redirects, error handling, invalidation).

Prerendering

`ls-tanstack-start/vite.config.ts`

```
tanstackStart({
  prerender: {
    enabled: true,
    crawlLinks: true, // follow links and prerender those too
    filter: ({ path }) =>
      ["/ssr_prerendered", "/ssr_awaited_prerendered"].includes(path),
  },
});
```

- **Off by default** — the opposite of Astro.
- At build, Start renders the chosen routes to static HTML.

Prerendering — the catch

`ls-tanstack-start/src/actions/index.ts`

```
import { staticFunctionMiddleware } from "@tanstack/start-static-server-functions";

export const getNavTime = createServerFn({ method: "GET" })
  .middleware([staticFunctionMiddleware]) // result baked at build time
  .handler(async () => ({ timeStr: new Date().toLocaleTimeString() }));
```

- Prerendering the *page* is not enough — the **server function results** must be frozen too.
- Otherwise the static HTML re-fetches on hydration and the timestamp jumps.

Streaming a slow part

`ls-tanstack-start/src/routes/ssr_awaited.tsx`

```
loader: (async () => ({
  nav: await getNavTime(),
  greetingPromise: getGreeting(), // NOT awaited → sent as a promise
}),
  (
    <Suspense fallback={<span aria-busy="true"></span>}>
      <Await promise={state.greetingPromise}>
        {(greeting) => (
          <Greeting browser={greeting.browser} timeStr={greeting.timeStr} />
        )}
      </Await>
    </Suspense>
  ));
```

- Start's answer to `server:defer` / PPR: **don't await it in the loader**.

TanStack Start summary

- The **router** is the framework; SSR is a feature bolted on top of a real SPA router.
- One component model; the boundary is `createServerFn`.
- End-to-end type safety: routes, links, loader data, server-fn input & output.
- Nothing is static unless you ask.
- Mental model: **a SPA that can also render on the server**.

A better framing than MPA vs SPA

> What does the framework **assume by default**, and what has to be **specially marked** to deviate?

- `Astro` — assumes **nothing runs**. Static HTML at build. Everything else opts in.
- `Next.js` — assumes the **server runs it fresh, per request**. Static is earned as an optimisation.
- `TanStack Start` — assumes the **server runs it, then hands the whole tree to the client**.

The "exception" axis

	Astro	Next.js	TanStack Start
Default	build-time static	per-request server	per-request server → SPA
Default output	static HTML, 0 JS	server-rendered HTML	server-rendered SPA shell
The exception	server island / client island	static generation / `"use client"`	static server function

> The exception is what you reach for to **break out of the default mode**.

What has to justify itself?

	What needs no justification	What needs a directive
Astro	static HTML	anything server-per-request, anything interactive
Next.js	server rendering every request	static pages, client-side interactivity
TanStack Start	SSR + SPA hydration	frozen / prerendered pages

Default execution unit

> At what granularity does the default mode apply?

	Default execution unit
Astro	per component (build time)
Next.js	per component (request time)
TanStack Start	per **route** (request time, then hydrates whole tree)

- Astro and Next both isolate at the **component** — islands sit next to each other with different timing.
- Start treats the **whole route** as one server execution, then rehydrates all of it into the SPA.

Static opt-in unit

> At what granularity do you opt **out** of the per-request default?

	Static opt-in unit
Astro	per component (`server:defer` / `client:*` to **add** dynamism to a static default)
Next.js	file / route (`"use client"`, `force-static`, `generateStaticParams`)
TanStack Start	**individual server function** (`staticFunctionMiddleware`)

Start's granularity is finer than its own route

```
// src/routes/index.tsx — one route, mixed timing
loader: async () => {
  const nav = await getNavTime();          // staticFunctionMiddleware → baked at build
  const greeting = await getGreeting(); // plain server fn → live per-request
  return { nav, greeting };
},
```

- `getNavTime` is frozen; `getGreeting` stays live — **on the same route**.
- In Next, the equivalent switch (`force-static`) is a file-level export that locks the whole route.
- Start's static unit is **finer** than the route default that contains it.

The ordering that matters

Astro	Next.js	TanStack Start
build-time first	server-first, but provably static	server-first, then client-owned

> **Astro**: nothing runs unless you say so.
>
> **Next.js**: the server runs it, but tries to prove it didn't have to.
>
> **Start**: the server runs it, then gives everything to the client.

Side by side — the same four ideas

Concept	Astro	Next.js	TanStack Start
Route	`src/pages/index.astro`	`src/app/page.tsx`	`src/routes/index.tsx`
Server-only render	`.astro` component	Server Component	server function (`createServerFn`)
Interactivity	`<C client:load />`	`"use client"`	every component
Client → server call	`actions.foo()`	`"use server"` fn	`useServerFn(foo)`

Side by side — rendering defaults

	Astro	Next.js	TanStack Start
Default output	**static**	static, dynamic where needed	**SSR per request**
Opt out	`prerender = false`	dynamic API usage	`prerender: { enabled: true }`
Deferred part	`server:defer` island	`<Suspense>` + PPR	unawaited promise + `<Await>`
JS shipped by default	**none**	client components only	**the whole app**

Side by side — the client ↔ server call

	Astro	Next.js	TanStack Start
Declared by	`defineAction`	`"use server"`	`createServerFn`
Input validation	`input: z.object()` ✅	none ❌	`.validator()` ✅
Errors	`{ data, error }`	`throw`	`throw`
HTTP method	POST	POST	you choose

A fifth axis — cacheability

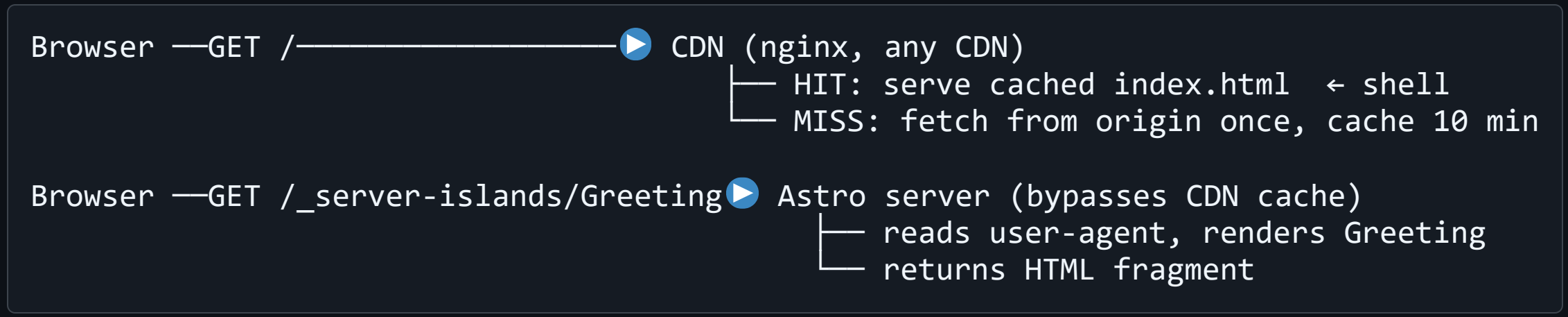
> The three frameworks ship **very different things** on the wire. That determines what a CDN can cache, without any app changes.

- **Astro** — two independent HTTP requests: shell (static) + island (dynamic).
- **Next.js PPR** — one chunked HTTP response: static shell + streamed holes in the same connection.
- **TanStack Start** — one per-request response: no static/dynamic split in the transport at all.

Server-side caching

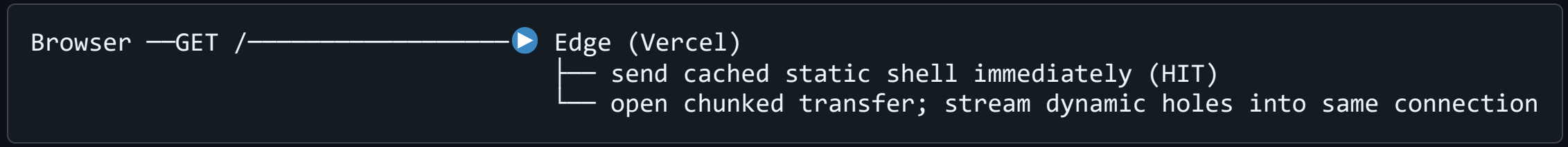
	What's cacheable	Cached by	Mechanism
Astro	shell + island, independently	any CDN incl. nginx	two separate HTTP GETs
Next.js	shell (in theory)	streaming-aware edge (Vercel)	one chunked response, shell + streamed holes
TanStack Start	only `staticFunctionMiddleware` output	build output / CDN	no shell; page is either fully live or frozen at build

Astro — two requests, two caches



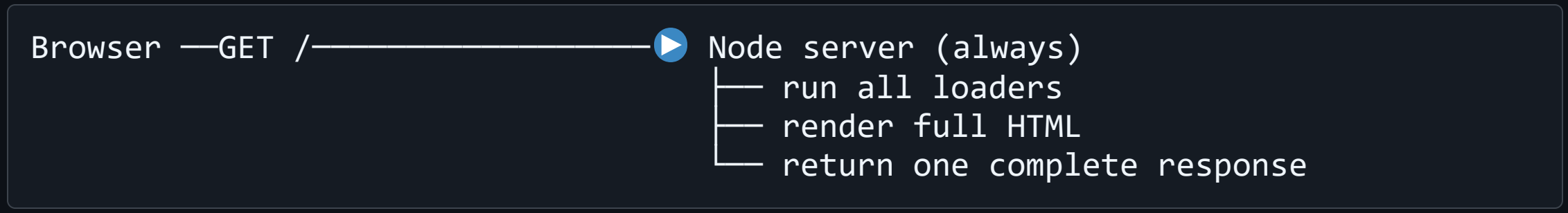
- The shell (`index.html`) and the island (`/_server-islands/Greeting`) are **different URLs** with different caching rules.
- nginx (or any reverse proxy) caches the shell the same way it caches any static file — no streaming support needed.
- The island can also be cached independently (e.g. 10s TTL) if it is not truly per-user.

Next.js PPR — one response, shared pipe



- The capability is real — the framework supports it.
- But shell + dynamic content share **one HTTP response**. nginx has no concept of "cache part of this response and pass through the rest of the same connection."
- **This is not a missing config option** — it is a genuine feature gap between generic reverse proxies and PPR-aware edge infrastructure built into Vercel.

TanStack Start — no middle tier



- There is **no shell/dynamic split** in the response — one document, generated fresh, every time.
- The only escape: pull data out of the per-request path via `staticFunctionMiddleware` — but that freezes at the individual function level.
- **Two modes, nothing in between:** fully live per-request, or function result frozen at build.

A sixth axis — client navigation caching

> This is separate from server caching. It governs what happens **after** the page is in the browser.

	Client-side navigation caching
Astro	None — MPA by default; each nav is a fresh full page load
Next.js	Router Cache for prefetched segments (implicit, version-dependent)
TanStack Start	Explicit SWR cache in the router: `staleTime`, `gcTime`, `preloadStaleTime` — configurable per route

> Start is the only one of the three that also gives you a fully explicit, tunable cache for what happens **after** the page arrives.

Start's router cache — three timers

```
export const Route = createFileRoute("/")({
  loader: () => getGreeting(),
  staleTime: 60_000, // navigating back within 60s reuses cached data
  preloadStaleTime: 10_000, // hover re-preloads if older than 10s
});
```

- `preloadStaleTime` — governs **hover** (prefetch trigger). Default: 30s.
- `staleTime` — governs **click** (navigation trigger). Default: 0 (always refetch).
- `gcTime` — how long cached data is kept in memory after the route is left.

> `staleTime: 0` is why every navigation reruns the loader even when the page was prerendered.

Why Start's client cache exists

- Astro and Next treat "the app is a persistent client-side router" as the exception.
  - Astro: full page reload per navigation by default.
  - Next: implicit router cache, behaviour changes between versions.
- Start treats the client router as the **primitive** — SSR is the server delivering the first render *to* the SPA.
- So Start is the only framework here with a fully documented, per-route, explicitly configurable client cache — because it has to be: the SPA never stops running.

A seventh axis — DX and mental models

> All three frameworks require you to learn something new. The question is **what kind of new**.

- `Astro` — keeps React completely separate from its own templating layer.
- `Next.js` — makes ordinary-looking React files behave differently based on invisible directives.
- `TanStack Start` — replaces invisible magic with real, callable JavaScript functions.

> **The single best DX lens:** is the framework's magic **implicit** (a convention you must memorize) or **explicit** (a function you can see, name, and trace)?

Astro DX — two familiar things, kept separate

- The `.astro` component is closer to a classic template language (Handlebars, EJS, Pug) than to React.
  - `---` fence = data fetching. Template below = HTML output. No hooks, no lifecycle.
  - Fits your intuition immediately because it *is* the old-school model.
- The React island is **100% ordinary React** — because it is a fully separate render tree with its own bundle.
  - No new React concepts. `useState`, `useEffect`, callbacks — all work as expected.
- **The trade-off:** islands are separate trees. They don't share React state or context with the Astro shell or with each other.
  - Astro's answer: external stores (`nanostores`) rather than React-native

Next.js DX — familiar syntax, unfamiliar rules

- You cannot `import` a Server Component inside a `'use client'` file.
- The escape hatch is **composition-by-children** — not a higher-order component pattern:

```
// page.tsx (Server Component)
<ClientWrapper>
  <ServerChild /> {/* rendered on server, passed as a resolved slot */}
</ClientWrapper>
```

- `ClientWrapper` never imports `ServerChild`; it just renders `{children}`.
  - The server resolved `ServerChild` before handing anything to the client tree.
  - Real pattern, but nothing in plain React trains you to think of `children` as a boundary-crossing mechanism.

Next.js DX — the root cause

> Why do both "`async` component / no hooks" and "can't import Server Component" feel foreign at the same time?

- Next repurposes **ordinary-looking React syntax** to mean something environment-dependent.
- A string literal at the top of the file (`'use client'` / `'use server'`) is intercepted by the compiler and silently changes what the file is allowed to do and where it runs.
- It looks identical to the React you know. It behaves differently.

> **Astro's unfamiliarity** is a new file type with new syntax → easy to see.
> **Next's unfamiliarity** is the same file type, same syntax, different semantics → harder to spot.

TanStack Start DX — explicit over implicit

- `createServerFn(...).handler(...)` is a **real function** you call, wrap, compose, and pass around.
- No magic string. No compiler rewriting the meaning of a component file.
- The server/client boundary is drawn by **which function you're inside**, not by a hidden annotation on the file.

```
export const getGreeting = createServerFn({ method: "GET" })
  .handler(async () => { ... });      // server boundary is here, in the call chain
```

- For someone from an SPA background: this is just "a hook that calls an API, with the network boundary handled for you."
- TanStack Router and Query are libraries SPA developers already have muscle memory for — Start reuses them rather than replacing them.

Implicit vs. explicit magic

	What you must learn	Magic type
Astro	`.astro` syntax, `client:*` directives, nanostores for cross-island state	Explicit — different file extension, tag-level directives you can see
Next.js	Server Component rules, `children` as boundary, `'use client'` / `'use server'` semantics	Implicit — same React syntax, compiler changes meaning invisibly
TanStack Start	`createServerFn` builder chain, loader/staleTime model	Explicit — real function calls you can read, name, and trace

> The throughline: Astro keeps React separate. Next changes React. Start

Which one?

Use case	Pick
Content, docs, marketing, blogs	`Astro`
Mostly-static pages with a few interactive corners	`Astro`
Server-heavy app, big team, Vercel-shaped deploy	`Next.js`
Dashboards, editors, anything behind a login	`TanStack Start`
You already love TanStack Query/Router	`TanStack Start`

> The Todo app is ~130 lines in **all three**. The difference is not the code — it's **what ships**.